

Bug Documentation

TypeScript — Input Type vs. DB Type Mismatch

Field	Details
Bug Name	Type Mismatch — Input Type vs. DB/Output Type
TypeScript Term	Assignability Error
Severity	Compile-time Error (blocks build)
Project Context	Next.js + Prisma + Zod + React Hook Form
File Affected	src/lib/actions/product.actions.ts

Error Message

```
Argument of type '{ name: string; slug: string; category: string; brand: string; description: string; stock: number; images: string[]; banner: string | null; price: string; isFeatured?: boolean | undefined; }' is not assignable to parameter of type 'Product'.
Type '...' is missing the following properties from type '{ id: string; rating: string; createdAt: Date; }': id, rating, createdAt
```

Root Cause

The `createProduct()` server action was typed to accept `Product` — the full database type that includes auto-generated fields (`id`, `rating`, `createdAt`). However, the form only collects input fields defined in `insertProductSchema`. The database generates the missing fields automatically — they should never come from the frontend.

Type Definitions

`insertProductSchema` (Zod — form input fields only):

```
const insertProductSchema = z.object({ name, slug, category, brand, description, stock, images, isFeatured, banner, price })
```

`Product` type (DB type — `insertSchema` + generated fields):

```
type Product = z.infer<typeof insertProductSchema> & { id: string // generated by DB
rating: string // generated by DB createdAt: Date // generated by DB }
```

The Bug

```
// product.actions.ts – WRONG export async function createProduct(data: Product) { ... }
```

The Fix

```
// product.actions.ts – CORRECT export async function createProduct( data: z.infer<typeof  
insertProductSchema> ) { ... }
```

The action accepts only the insert schema. The DB generates `id`, `rating`, and `createdAt` automatically via Prisma.

How It Was Debugged

- 1 Identified the error pointed to missing fields: `id`, `rating`, `createdAt`
- 2 Checked the `createProduct` function signature in the actions file
- 3 Used `grep` to confirm the file on disk still had the old type: `grep -n "createProduct" src/lib/actions/product.actions.ts`
- 4 Changed parameter type from `Product` to `z.infer`
- 5 Restarted TypeScript server in VS Code to clear stale type cache

Lesson Learned

Always use the **input/insert type** (what the form sends) as the parameter for create actions, not the full **DB/output type** (what the database returns). DB-generated fields like `id`, `rating`, and `createdAt` should never be expected from the frontend.